ROSP (Recent Open Source Project) Lab Report

Student Name: Gaurav Salekar	**GitHub:** github.com/gaurav-about
Subject: ROSP Lab	**Project:** Google Flutter (flutter/flutter)
Repository: flutter-rosp-contribution	**Pull Request:** Merged #2

1. Problem Statement / Issue Identification

Issue Title: [material] Badge widget ignores custom offset when label is null

Affected Components:

- **Source File:** packages/flutter/lib/src/material/badge.dart
- **Test File:** packages/flutter/test/material/badge_test.dart

Description:

In Flutter Material 3, the Badge widget decorates an icon or widget with a small status dot or numerical label. When label != null (large badge), the badge displays label text and respects the offset parameter. However, when label == null (small badge dot), the constructor still accepts offset, but the internal implementation forced Offset.zero, discarding the developer's requested position entirely.

```
// packages/flutter/lib/src/material/badge.dart (Line 242)
Positioned.fill(
  child: _Badge(
    alignment: effectiveAlignment,
    offset: hasLabel ? effectiveOffset : Offset.zero, // <-- BUG: Always Offset.zero!
    hasLabel: hasLabel,
    widthOffset: effectiveWidthOffset,
    textDirection: textDirection,
    child: badge,
  ),
),
```

2. Root Cause Analysis

1. effectiveOffset was calculated using (offset ?? badgeTheme.offset ?? defaultOffset) + const Offset(0, 8) for large badges.
2. For small badges (hasLabel == false), the ternary condition passed Offset.zero directly, entirely ignoring caller-specified offsets.
3. The render pipeline in _RenderBadge already applies offset: Offset badgeLocation = offset + resolvedAlignment.alongOffset(...). Because Offset.zero was passed, the calculated location never shifted.

3. The Implementation / Bug Fix

We modified Badge.build so that when hasLabel is false, it uses (offset ?? badgeTheme.offset ?? Offset.zero).

```
--- a/packages/flutter/lib/src/material/badge.dart
+++ b/packages/flutter/lib/src/material/badge.dart
@@ -240,7 +240,7 @@ class Badge extends StatelessWidget {
         Positioned.fill(
           child: _Badge(
             alignment: effectiveAlignment,
-            offset: hasLabel ? effectiveOffset : Offset.zero,
+            offset: hasLabel ? effectiveOffset : (offset ?? badgeTheme.offset ?? Offset.zero),
             hasLabel: hasLabel,
             widthOffset: effectiveWidthOffset,
             textDirection: textDirection,
             child: badge,
           ),
         ),
```

4. Automated Unit Test Verification

We added a comprehensive unit test in packages/flutter/test/material/badge_test.dart:

```
testWidgets('Small Badge respects custom offset', (WidgetTester tester) async {
  final theme = ThemeData();
  await tester.pumpWidget(
    MaterialApp(
      theme: theme,
      home: const Align(
        alignment: Alignment.topLeft,
        child: Badge(offset: Offset(2, -2), child: Icon(Icons.add)),
      ),
    ),
  );
  // icon width = 24, smallSize = 6 -> offset (2, -2) yields rect: (20, -2, 26, 4)
  final RenderBox box = tester.renderObject(find.byType(Badge));
  expect(box, paints..rrect(
    rrect: RRect.fromLTRBR(20, -2, 26, 4, const Radius.circular(3)),
    color: theme.colorScheme.error,
  ));
});
```

Test Results:

- **Before Fix:** Failed with Expected: RRect(20.0, -2.0, 26.0, 4.0), Actual: RRect(18.0, 0.0, 24.0, 6.0)
- **After Fix:** 00:01 +1: Small Badge respects custom offset - All tests passed!

5. GitHub Version Control & Pull Request Verification

- **Repository:** https://github.com/gaurav-about/flutter-rosp-contribution
- **Pull Request:** https://github.com/gaurav-about/flutter-rosp-contribution/pull/2
- **PR Title:** fix(material): support custom offset for small Badge without label #2
- **Status:** **Merged** (Merged 1 commit into master from fix/small-badge-custom-offset)
- **Files Changed:** 2 files (badge.dart, badge_test.dart) with +36 additions, -4 deletions
- **Author / Contributor:** Gaurav Salekar (gaurav-about)

6. Conclusion

The open-source contribution successfully resolved an edge-case layout defect in Google Flutter's Material library. The bug was diagnosed at the render layout boundary, resolved with full backward compatibility, verified via automated unit tests, committed under standard conventional commit conventions, and officially merged via GitHub Pull Request.